

# ZIQIAN WANG

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## EDUCATION

University of Michigan, Ann Arbor

Bachelor of Science and Engineering in Computer Science

Fall 2026 (Incoming)

Shanghai Jiao Tong University

Bachelor of Engineering in Electrical and Computer Engineering

Sep 2024 – Present

GPA: 3.95/4.0

## RESEARCH EXPERIENCE

AgiBot, Shanghai

Research Intern

May 2026 – Present

Shanghai, China

- Scaling up prior **off-policy RL** fine-tuning research to fleet scale with abundant compute and a **real-robot fleet**.
- Building a **continual-learning and self-evolution** system over the real-robot fleet so that deployed policies keep improving autonomously from continuously collected real-world interaction data.
- Introducing a **human-in-the-loop intervention** mechanism that folds real-time human corrections into the data and training loop to improve safety, data quality, and sample efficiency.

ScaleLab, Shanghai Jiao Tong University

Research Intern (Advisor: Prof. Yao Mu)

Oct 2025 – Present

Shanghai, China

- First-author paper: “**Q-VGM: Q-Guided Value-Gradient Matching for Flow-Matching VLA Policies**”; proposed an **off-policy RL** fine-tuning method for **Pi0.5**, a 3B-parameter flow-matching Vision-Language-Action (VLA) policy.
- Following a **few-shot-SFT then learn-from-experience** paradigm, raised average success from **75.0% to 92.5%** on **LIBERO**, **76.4% to 87.2%** on **RoboTwin 2.0**, and **40.0% to 67.5%** on two real-robot tabletop tasks, outperforming all same-backbone, same-critic baselines.

Intelligent Vehicles Lab, Shanghai Jiao Tong University

Research Assistant (Advisor: Prof. Qin Tong)

Apr 2025 – Oct 2025

Shanghai, China

- Co-developed **Scene2Gym**, a framework that converts real-world scenes into interactive, metric-consistent **3DGS** digital twins for embodied policy learning. Second-author paper **accepted at IROS 2026**.
- Led the **RL and imitation learning policy training** pipeline within the Scene2Gym framework, developing and tuning PPO-based and behavior-cloning policies for contact-rich manipulation tasks in photorealistic simulated environments.
- Achieved **56%** zero-shot sim-to-real transfer for visual RL policies and near-demonstration-level performance for imitation learning, demonstrating the importance of visual fidelity and perception-action consistency for transferable embodied learning.

## PUBLICATIONS

**Z. Wang**, et al., “Q-VGM: Q-Guided Value-Gradient Matching for Flow-Matching VLA Policies,” *Under Review*. (**First Author**) [arXiv] (preliminary version; additional experiments in progress)

Author, **Z. Wang** et al., “Scene2Gym: Transforming Real-World Scenes into Interactive Training Environments for Robot Learning,” *Accepted at IROS 2026*. (Second Author)

## RESEARCH INTERESTS

I am interested in building embodied agents that interact with the physical world. Specifically, I aim to explore how reinforcement learning, foundation models, and world models can be combined to build generalist robots.

## TECHNICAL SKILLS

**Programming**  
**Frameworks**

Python, C++ (NOIP First Prize), MATLAB  
PyTorch, MuJoCo, Isaac Gym, ROS, Weights & Biases, Claude Code